



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES	DATE	10 OCTOBER 2022
GRADE	11	MARKS	50
EXAMINER	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
NAME		DURATION	50 MINUTES
CLASS			

INSTRUCTIONS:

1. This paper consists of 6 pages. Please check that your question paper is complete.
 2. Answer all questions on the folio paper provided.
 3. Write in blue or black pen only.
 4. Read all questions carefully and answer in as much detail as possible.
 5. All diagrams should have a clear heading.
 6. It is in your own interest to write legibly and to present your work neatly.
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QUESTION	POSSIBLE MARK	MARK ACHIEVED
1	10	
2	11	
3	19	
4	10	
TOTAL: 50		

Question 1:

1.1. Various possibilities are suggested as answers to the following questions. Indicate the correct answer by writing the correct letter from the corresponding question on the folio paper provided.

1.1.1. The part of the brain that controls breathing and the heartbeat is the

- A. Medulla oblongata
- B. cerbrum
- C. cerebellum
- D. corpus callosum

1.1.2. The sympathetic nervous:

- A. Is antagonistic to the parasympathetic nervous system
- B. Speeds up the heart rate
- C. Works in response to the hormone adrenalin
- D. All of the above

1.1.3. The following events occur during an action potential:

- i. Repolarization occurs
- ii. Threshold potential is reached
- iii. The neuron is at resting potential
- iv. Depolarization occurs

Select the correct order in which these events occur:

- A. iii, ii, iv, i
- B. i, ii, iii, iv
- C. iii, iv, ii, i
- D. ii, i, iv, iii

1.1.4. The nerves that branch directly from the brain stem and transmit nerve signals to and from the eyes, ears, mouth, face and scalp are known as...

- A. Meninges
- B. Spinal nerves
- C. Cerebral ganglia
- D. Cranial nerves

1.1.5. What structure is responsible for transmitting signals from the peripheral nervous system to the central nervous system?

- A. An interneuron
- B. A sensory neuron
- C. A motor neuron
- D. A ganglion

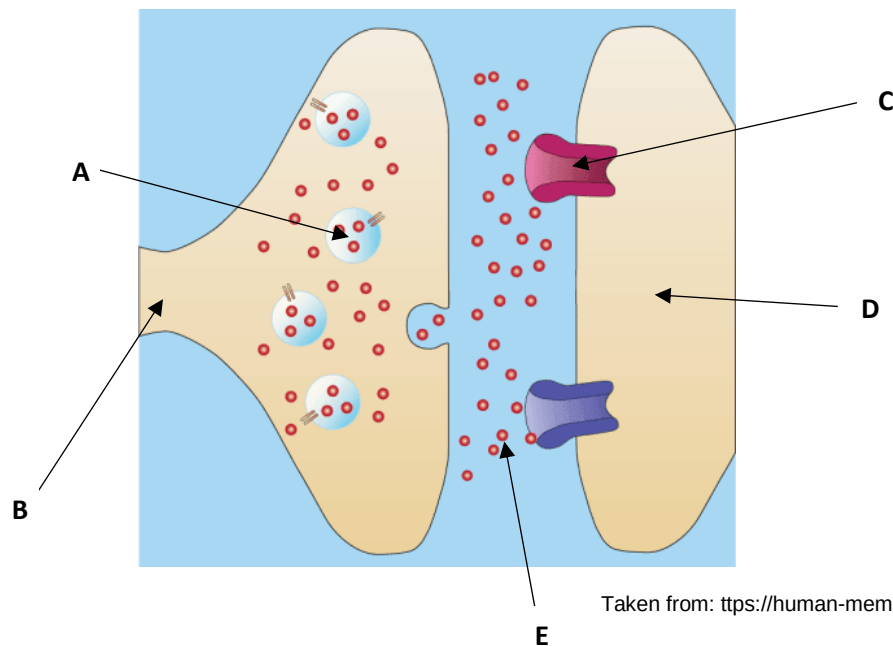
(5)

- 1.2. Complete each of the following statements by writing down ONLY the missing word on your answer paper.
- 1.2.1. The traffic light turns green and the driver steps on the gas pedal to make the car move forward. The green light acted as a _____ that caused a response in the driver.
- 1.2.2. The type of neuron known as _____ neurons pickup stimuli from the external or internal environment and convert those stimuli to nerve impulses.
- 1.2.3. A muscle contracts in response to an impulse carried by the type of neuron known as a _____ neuron.
- 1.2.4. If you accidentally touch a hot pan, you immediately jerk your hand away without thinking, before you even feel pain. This type of response is known as a _____.
- 1.2.5. One function of the nervous system is to maintain _____, or stability within the body. (5)

[10]

Question 2:

Study the diagram below and answer the questions that follow.



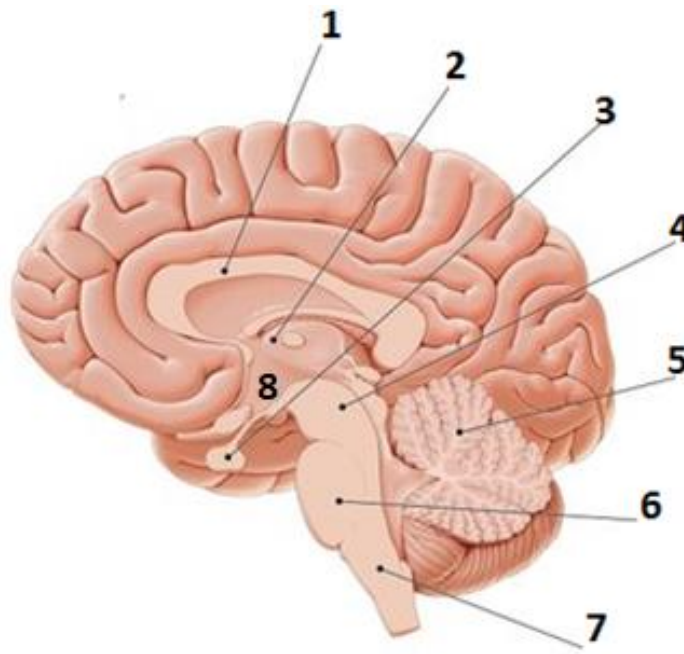
- 2.1. Provide labels for structures B and D. (2)

- 2.2. What is the function of the chemical substances shown at E? (2)
- 2.3. Why is the above structure important in the functioning of the nervous system? (2)
- 2.4. Using the NAMES of the structures above describe how an impulse travels from one neuron to another at a synapse. (5)

[11]

Question 3

Study the diagram below and answer the questions that follow.



Taken from: <https://www.kenhub.com/en/library/learning-strategies>

- 3.1. Provide labels for structures 1 and 5. (2)
- 3.2. Provide TWO functions structure 6. (2)
- 3.3. How is the above structure protected? (2)
- 3.4. Choose the key responses (A - D) that best correspond to the descriptions provided in the following statements. Insert only the appropriate letter on your folio paper. (4)

A. Autonomic nervous system

C. Peripheral nervous system

B. Central nervous system

D. Somatic nervous system

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- The diagram illustrates a reflex arc. On the left, a hand is shown with an arrow pointing to the **SKIN**. A line representing the sensory neuron leads from the skin to a small circle labeled **A** (the cell body). From **A**, the axon leads to a point labeled **B** (the axon hillock). The axon then enters the **SPINAL CORD**, which is shown as a large, oval structure. Inside the spinal cord, the axon terminates at a point labeled **C** (the synapse). From **C**, the axon of the motor neuron leads to a point labeled **D** (the cell body). The axon then leads to a point labeled **E** (the axon hillock). The axon exits the spinal cord and leads to a hand with an arrow pointing to the **MUSCLE**. A curved arrow labeled **F** indicates the direction of the impulse flow from the skin to the muscle.

- 3.5.1. What is the difference between afferent neurons and efferent neurons (2)
- 3.5.2. Provide the name of the neuron indicated by B and F. (1)
- 3.5.3. What is the effector indicated in the diagram above? (1)
- 3.5.4. If a nerve impulse travels at 76 m/s, how long will it take the reflex arc to be completed if the distance between the area of touch and the spinal cord is 1.2 meters? (round of to two decimal places) (3)
- 3.5.5. Explain what will happen to this person if structure F were to be cut off and the stimulus received was a pin prick? (2)

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Question 4

4.1. Name the part of the brain responsible for coordinating or controlling each activity or condition listed below. (5)

4.1.1. Tasting a hamburger.

4.1.2. Solving a maths problem.

4.1.3. Kicking a soccer ball.

4.1.4. Rate of breathing.

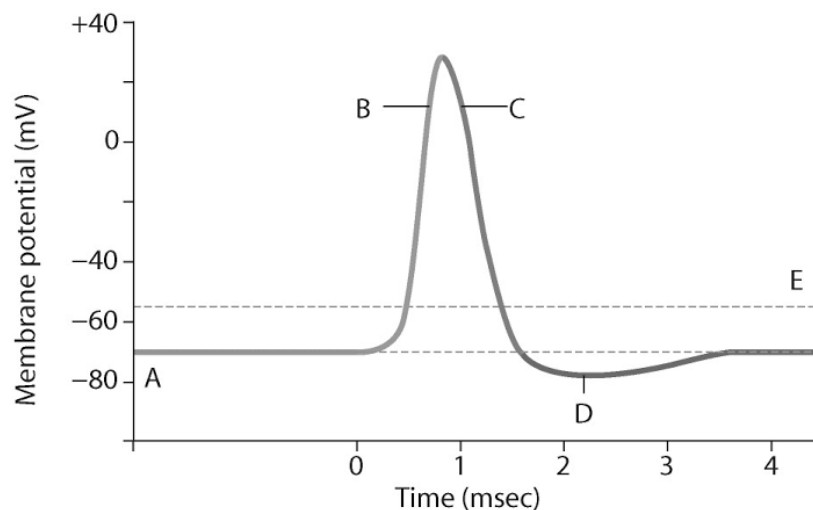
4.1.5. Looking for your cat.

4.2. The diagram below resembles an action potential. For the explanations A-E below write on your folio paper P, D or R.

P- if it describes a polarized neuron

D- if it describes a neuron that is depolarizing

R- if it describes a neuron that is repolarizing



A. Potassium is exiting the neuron.

B. Sodium is entering the neuron.

C. Threshold potential is reached.

D. The neuron is at rest.

E. The neuron is getting more positive outside.

(5)

[10]

TOTAL: 50 MARKS